

Module 5

Two-Port Networks and Network Graphs

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Relationship between parameters

For any two sets of parameters, we can relate one set to the other set.

Given the z parameters, let us obtain the y parameters.

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{z}] \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{z}]^{-1} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

Also

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = [\mathbf{y}] \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

Comparing,

$$[\mathbf{y}] = [\mathbf{z}]^{-1}$$

$$\begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} = \frac{\begin{bmatrix} \mathbf{z}_{22} & -\mathbf{z}_{12} \\ -\mathbf{z}_{21} & \mathbf{z}_{11} \end{bmatrix}}{\Delta_z}$$

$$\mathbf{y}_{11} = \frac{\mathbf{z}_{22}}{\Delta_z}, \quad \mathbf{y}_{12} = -\frac{\mathbf{z}_{12}}{\Delta_z}, \quad \mathbf{y}_{21} = -\frac{\mathbf{z}_{21}}{\Delta_z}, \quad \mathbf{y}_{22} = \frac{\mathbf{z}_{11}}{\Delta_z}$$

Relationship between parameters

Given the z parameters, let us obtain the h parameters.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2 \quad \text{---1}$$

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2 \quad \text{---2}$$

From 2

$$\mathbf{I}_2 = -\frac{\mathbf{z}_{21}}{\mathbf{z}_{22}}\mathbf{I}_1 + \frac{1}{\mathbf{z}_{22}}\mathbf{V}_2 \quad \text{---3}$$

Put 3 in 1

$$\mathbf{V}_1 = \frac{\mathbf{z}_{11}\mathbf{z}_{22} - \mathbf{z}_{12}\mathbf{z}_{21}}{\mathbf{z}_{22}}\mathbf{I}_1 + \frac{\mathbf{z}_{12}}{\mathbf{z}_{22}}\mathbf{V}_2 \quad \text{---4}$$

From 3 and 4

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \frac{\Delta_z}{\mathbf{z}_{22}} & \frac{\mathbf{z}_{12}}{\mathbf{z}_{22}} \\ -\frac{\mathbf{z}_{21}}{\mathbf{z}_{22}} & \frac{1}{\mathbf{z}_{22}} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

$$\mathbf{h}_{11} = \frac{\Delta_z}{\mathbf{z}_{22}}, \quad \mathbf{h}_{12} = \frac{\mathbf{z}_{12}}{\mathbf{z}_{22}}, \quad \mathbf{h}_{21} = -\frac{\mathbf{z}_{21}}{\mathbf{z}_{22}}, \quad \mathbf{h}_{22} = \frac{1}{\mathbf{z}_{22}}$$

Interconnection of networks

A large, complex network may be divided into sub networks for the purposes of analysis and design.

The sub networks are modeled as two port networks, interconnected to form the original network.

The two-port networks may therefore be regarded as building blocks that can be interconnected to form a complex network.

The interconnection can be in series, in parallel, or in cascade.

For example, when the networks are in **series**, their **individual z parameters add** up to give the z parameters of the larger network.

When they are in **parallel**, their **individual y parameters add** up to give the y parameters of the larger network.

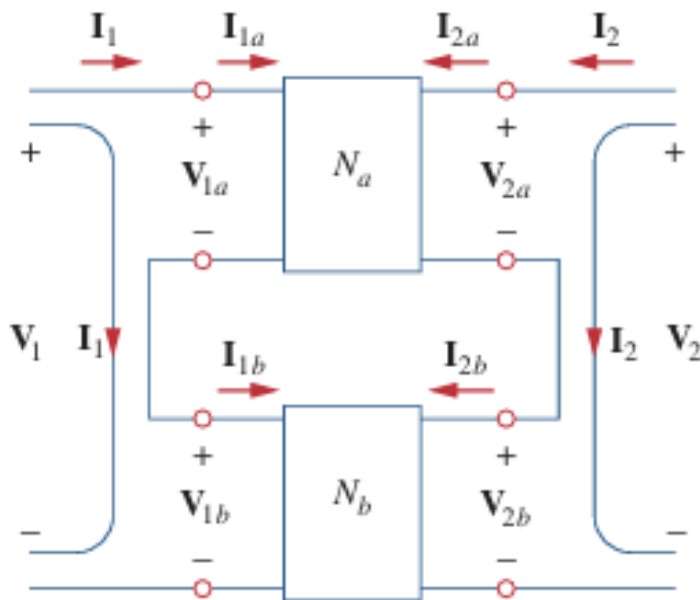
When they are **cascaded**, their **individual transmission parameters can be multiplied** together to get the transmission parameters of the larger network.

Interconnection of networks (series)

Consider the series connection of two two-port network.

The networks are regarded as being in series because their input currents are the same and their voltages add.

In addition, each network has a common reference, and when the circuits are placed in series, the common reference points of each circuit are connected together.



For network 'a'

$$V_{1a} = z_{11a}I_{1a} + z_{12a}I_{2a}$$

$$V_{2a} = z_{21a}I_{1a} + z_{22a}I_{2a}$$

For network 'b'

$$V_{1b} = z_{11b}I_{1b} + z_{12b}I_{2b}$$

$$V_{2b} = z_{21b}I_{1b} + z_{22b}I_{2b}$$

From the figure,

$$I_1 = I_{1a} = I_{1b}, \quad I_2 = I_{2a} = I_{2b}$$

and

$$V_1 = V_{1a} + V_{1b} = (z_{11a} + z_{11b})I_1 + (z_{12a} + z_{12b})I_2$$

$$V_2 = V_{2a} + V_{2b} = (z_{21a} + z_{21b})I_1 + (z_{22a} + z_{22b})I_2$$

Interconnection of networks (series)

Thus, the z parameters for the overall network are

$$\begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11a} + \mathbf{z}_{11b} & \mathbf{z}_{12a} + \mathbf{z}_{12b} \\ \mathbf{z}_{21a} + \mathbf{z}_{21b} & \mathbf{z}_{22a} + \mathbf{z}_{22b} \end{bmatrix}$$

$$[\mathbf{z}] = [\mathbf{z}_a] + [\mathbf{z}_b]$$

This can be extended to n networks in series.

If two two-port networks in the h -model, for example, are connected in series, we convert the h to z to and then apply $[\mathbf{z}] = [\mathbf{z}_a] + [\mathbf{z}_b]$

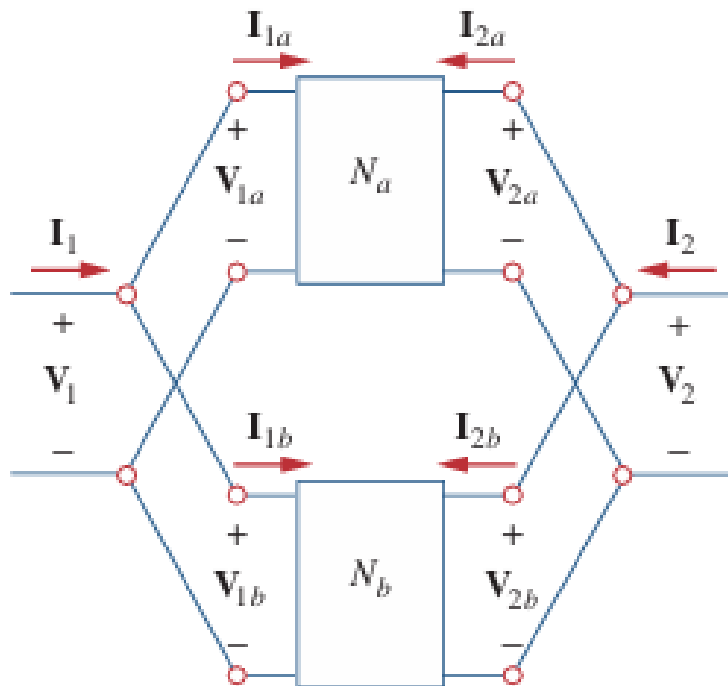
Then convert back to h .

Interconnection of networks (parallel)

Two two-port networks are in parallel when their port voltages are equal and the port currents of the larger network are the sums of the individual port currents.

Each circuit must have a common reference and when the networks are connected together, they must all have their common references tied together.

The parallel connection of two two-port networks is shown:



For network 'a'

$$\begin{aligned} I_{1a} &= y_{11a}V_{1a} + y_{12a}V_{2a} \\ I_{2a} &= y_{21a}V_{1a} + y_{22a}V_{2a} \end{aligned} \quad \text{---1}$$

For network 'b'

$$\begin{aligned} I_{1b} &= y_{11b}V_{1b} + y_{12b}V_{2b} \\ I_{2b} &= y_{21b}V_{1b} + y_{22b}V_{2b} \end{aligned} \quad \text{---2}$$

From the figure,

$$\begin{aligned} V_1 &= V_{1a} = V_{1b}, & V_2 &= V_{2a} = V_{2b} \\ I_1 &= I_{1a} + I_{1b}, & I_2 &= I_{2a} + I_{2b} \end{aligned} \quad \text{---3}$$

Interconnection of networks (parallel)

Substituting 1 and 2 in 3,

$$\mathbf{I}_1 = (\mathbf{y}_{11a} + \mathbf{y}_{11b})\mathbf{V}_1 + (\mathbf{y}_{12a} + \mathbf{y}_{12b})\mathbf{V}_2$$

$$\mathbf{I}_2 = (\mathbf{y}_{21a} + \mathbf{y}_{21b})\mathbf{V}_1 + (\mathbf{y}_{22a} + \mathbf{y}_{22b})\mathbf{V}_2$$

$$\begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11a} + \mathbf{y}_{11b} & \mathbf{y}_{12a} + \mathbf{y}_{12b} \\ \mathbf{y}_{21a} + \mathbf{y}_{21b} & \mathbf{y}_{22a} + \mathbf{y}_{22b} \end{bmatrix}$$

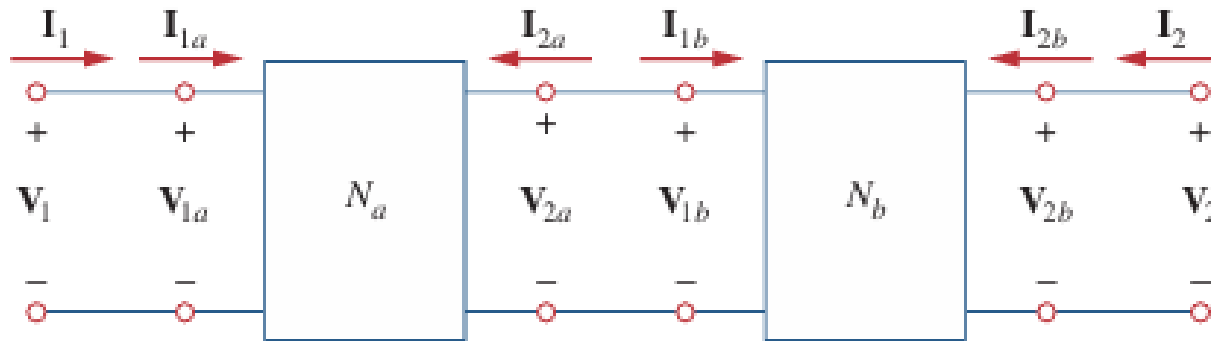
$$[\mathbf{y}] = [\mathbf{y}_a] + [\mathbf{y}_b]$$

Shows that the y parameters of the overall network are the sum of the y parameters of the individual networks.

Interconnection of networks (cascaded)

Two networks are said to be cascaded when the output of one is the input of the other.

The connection of two two-port networks in cascade is shown:



$$\begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix} = \begin{bmatrix} A_a & B_a \\ C_a & D_a \end{bmatrix} \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix} \quad \text{---1}$$

$$\begin{bmatrix} V_{1b} \\ I_{1b} \end{bmatrix} = \begin{bmatrix} A_b & B_b \\ C_b & D_b \end{bmatrix} \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix} \quad \text{---2}$$

From the figure,

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} V_{1a} \\ I_{1a} \end{bmatrix}, \quad \begin{bmatrix} V_{2a} \\ -I_{2a} \end{bmatrix} = \begin{bmatrix} V_{1b} \\ I_{1b} \end{bmatrix}, \quad \begin{bmatrix} V_{2b} \\ -I_{2b} \end{bmatrix} = \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \quad \text{---3}$$

Interconnection of networks (cascaded)

From 1 and 2 in 3,

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_a & \mathbf{B}_a \\ \mathbf{C}_a & \mathbf{D}_a \end{bmatrix} \begin{bmatrix} \mathbf{A}_b & \mathbf{B}_b \\ \mathbf{C}_b & \mathbf{D}_b \end{bmatrix} \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix}$$

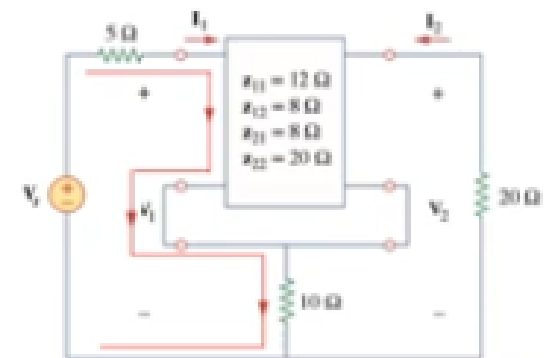
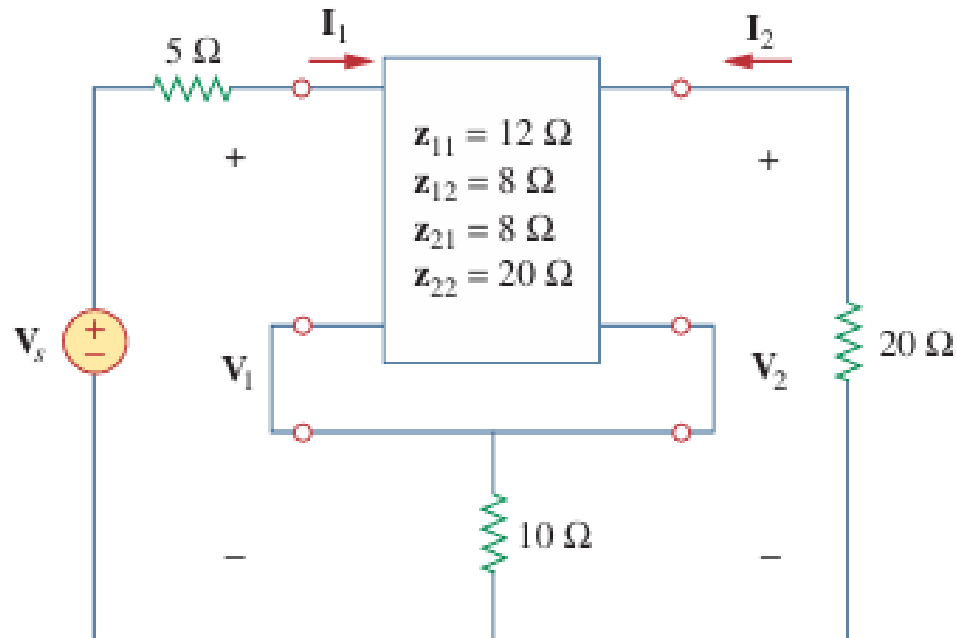
$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_a & \mathbf{B}_a \\ \mathbf{C}_a & \mathbf{D}_a \end{bmatrix} \begin{bmatrix} \mathbf{A}_b & \mathbf{B}_b \\ \mathbf{C}_b & \mathbf{D}_b \end{bmatrix}$$

$$\boxed{[\mathbf{T}] = [\mathbf{T}_a][\mathbf{T}_b]}$$

Thus, the transmission parameters for the overall network are the product of the transmission parameters for the individual transmission parameters.

Interconnection of networks

1. Evaluate V_o/V_s in the circuit.



For N_b $\mathbf{z}_{12b} = \mathbf{z}_{21b} = 10 = \mathbf{z}_{11b} = \mathbf{z}_{22b}$

$$[\mathbf{z}] = [\mathbf{z}_a] + [\mathbf{z}_b] = \begin{bmatrix} 12 & 8 \\ 8 & 20 \end{bmatrix} + \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix} = \begin{bmatrix} 22 & 18 \\ 18 & 30 \end{bmatrix}$$

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2 = 22\mathbf{I}_1 + 18\mathbf{I}_2 \quad \text{---1}$$

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2 = 18\mathbf{I}_1 + 30\mathbf{I}_2 \quad \text{---2}$$

Interconnection of networks

at the input port

$$\mathbf{V}_1 = \mathbf{V}_s - 5\mathbf{I}_1 \quad \text{---3}$$

at the output port

$$\mathbf{V}_2 = -20\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_2 = -\frac{\mathbf{V}_2}{20} \quad \text{---4}$$

Substituting 3 and 4 in 1,

$$\mathbf{V}_s - 5\mathbf{I}_1 = 22\mathbf{I}_1 - \frac{18}{20}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_s = 27\mathbf{I}_1 - 0.9\mathbf{V}_2 \quad \text{---5}$$

Substituting 4 in 2,

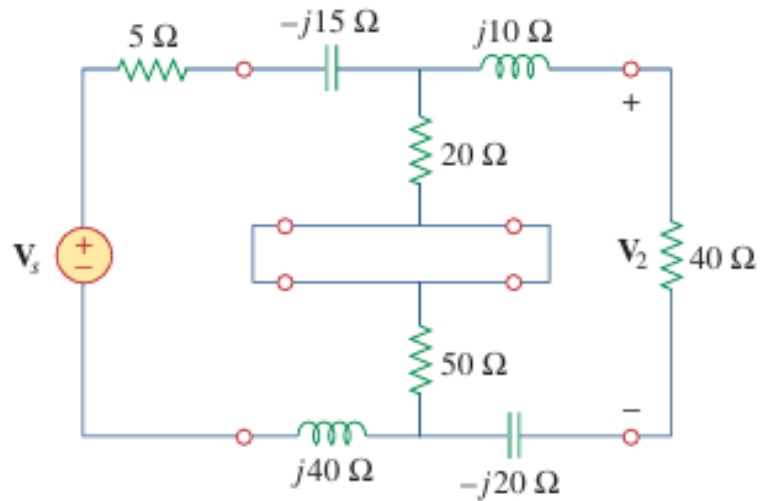
$$\mathbf{V}_2 = 18\mathbf{I}_1 - \frac{30}{20}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{I}_1 = \frac{2.5}{18}\mathbf{V}_2 \quad \text{---6}$$

Substituting 6 in 5,

$$\mathbf{V}_s = 27 \times \frac{2.5}{18}\mathbf{V}_2 - 0.9\mathbf{V}_2 = 2.85\mathbf{V}_2 \qquad \frac{\mathbf{V}_2}{\mathbf{V}_s} = \frac{1}{2.85} = 0.3509$$

Interconnection of networks

2. Evaluate V_2/V_s in the circuit.



Next class: Network Graphs